Lab 12: Fast LRC Decay

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Pre-lab Questions

Derive the relationship between the time constant in an LRC circuit and the half life of the voltage across the resistor.

From the general solution for the voltage across the resistor in an LRC circuit, we have:

$$V(t) = V_0 e^{-\frac{R}{2L}t} \cos(\omega t + \varphi),$$

where $\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$ is the angular frequency of the oscillation. The time constant τ is defined as the time it takes for the voltage to decay to $1/e$ of its initial value, which occurs when the exponential is $1/e$:

$$e^{-\frac{R}{2L}\tau} = \frac{1}{e} \quad \rightarrow \quad -\frac{R}{2L}\tau = -1 \quad \rightarrow \quad \tau = \frac{2L}{R}.$$

This allows us to rewrite the voltage as:

$$V(t) = V_0 e^{-t/\tau} \cos(\omega t + \varphi).$$

The half-life $t_{1/2}$ is defined as the time it takes for the voltage to decay to half of its initial value, which occurs when the exponential is 1/2:

$$e^{-t_{1/2}/\tau} = \frac{1}{2} \quad \rightarrow \quad -t_{1/2}/\tau = \ln\left(\frac{1}{2}\right) \quad \rightarrow \quad t_{1/2} = \tau \ln 2 = \frac{2L \ln 2}{R}.$$

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Part 1: Driven LRC Series Circuit

An LC circuit with an inductor of labeled inductance 22 mH and a capacitor of labeled capacitance 0.01 μF = 10 nF is driven by a function generator with a square wave output, with some internal resistance R.

Inductor value (measured)	$L \approx 22.8\,\text{mH}$
Capacitor value (measured)	$C \approx 9.6\,\text{nF}$
Total resistance (measured)	$R \approx 36.4\,\Omega + 50\,\Omega = 86.4\,\Omega$

It is known that the

	Theoretical	**Experimental**	**% Error**
Oscillation Frequency, f	**Half life, $t_{1/2}$**		

Conclusion

1. In part 1, you were asked to select an appropriate frequency for the square wave. How do you know that you have an appropriate frequency?
2. In part 3, how do you know when your circuit is underdamped, critically damped, and overdamped?
3. In part 3, what happens to the potential across the resistor as the resistance in the circuit is increased from its lowest to its highest value?
4. Why does the value of the resistor change the oscillation of the potential? Your answer should include references to the power input to the circuit from the generator, and the power dissipated in the resistor.